

DAY 01 · LAB

Size a Facility.

DURATION

90 minutes

TEAMS OF

Three

YOU NEED

Pen, paper, a calculator

STRUCTURE

Eight steps, one sum each

EVERY STEP HAS THE SAME THREE PARTS

Guide	The idea, in plain language, and the one number you need
Do	One sum. Write the answer in the box
Question	Something to think about before you move on

Each step uses the answer from the step before. Work in order, do not skip, and write every answer down where you can see it. If a number looks strange, fix it before you carry it forward.

YOUR BRIEF

Your university is building a research computing facility. That is all you have been told.

Servers	400
Each one draws	0.5 kW
Servers per rack	20

Everyone in the room designs this same facility, so you can compare answers. Your curveball comes at the end, and nobody else has yours.

RUN SHEET

0:00	This slide
0:05	Steps 1 to 8
0:45	Check your answers
0:50	Step 9, reliability
0:55	Step 10, your curveball
1:10	Write your card
1:20	Readouts

STEP 01 OF 08

How much power do the servers need?

GUIDE

A server is a small heater with a computer inside it. Nearly everything it draws from the wall comes back out as heat, which is the fact the whole lab rests on.

An ordinary one draws about **0.5 kW**. That is an average over a day, not the largest figure it could ever reach, and not the number printed on its power supply.

THE RULE

One server = 0.5 kW

DO

400 servers × 0.5 kW

=

KW

Call this the server power.

QUESTION

A domestic kettle draws about 2 kW.

How many kettles, all boiling at once and never switching off, is your room of servers?

STEP 02 OF 08

What else is in the room?

GUIDE

Servers do not sit alone. There are network switches moving data between them, storage holding the data, and the equipment that connects it all together.

Nobody photographs any of it, and all of it is on all the time. Add **10%** to what you just worked out.

THE RULE

Add 10% for switches and storage

DO

FROM STEP 01

server power + 10%

=

KW

This is your computing load. Circle it. Four of the remaining steps use it.

QUESTION

Suppose you got this number wrong by 10%.

Which of your later answers would also be wrong, and roughly by how much?

STEP 03 OF 08

How many racks do you need?

GUIDE

A rack is a standard steel cabinet, roughly the size and shape of a domestic fridge, with servers stacked inside it on rails.

Racks are all the same size everywhere in the world, which is why a server bought in one country fits a cabinet in another. About **20** ordinary servers fit in one.

THE RULE

20 servers per rack

DO

400 servers ÷ 20 per rack

=

RACKS

Call this your rack count.

QUESTION

You now have that many fridge-sized cabinets to put somewhere.

Sketch the room. Where do the cabinets go, where do people walk, and which side does the hot air come out of?

STEP 04 OF 08

Can you actually cool them?

GUIDE

This is the step that decides whether your design is real.

Air can only carry away so much heat. Fans and air conditioning stop being enough at roughly **15 kW per rack**. Above that you need contained aisles, or you cool the chips directly with liquid, which is a different and much more expensive building.

THE RULE

15 kW per rack is the air cooling ceiling

DO

FROM STEPS 02 AND 03

computing load ÷ rack count

=

KW PER RACK

Circle one: under 15 / over 15

QUESTION

You could halve your floor space by putting 40 servers in each rack instead of 20. Do that sum.

Why does nobody do this?

STEP 05 OF 08

What does the building draw?

GUIDE

All that electricity turns into heat, and removing heat takes more electricity again: chillers, pumps, fans, and losses in the power equipment itself.

For every 1 kW of computing, a typical building draws about **1.5 kW** in total. That ratio has a name, PUE, and it is the single number the industry uses to judge whether a building is any good. The best in the world are close to 1.1.

THE RULE

Building power = computing load × 1.5

DO

FROM STEP 02

computing load × 1.5

=

KW

This is your building power. Steps 7 and 8 need it.

QUESTION

Subtract your computing load from your building power. That difference is electricity you bought and did no computing with.

Name three things it is being spent on.

STEP 06 OF 08

How big a room is that?

GUIDE

A rack takes up about 0.7 m² of floor, but you cannot pack them tight. Engineers need to open both doors, and cold air must reach the front while hot air leaves the back. So allow **3 m²** per rack.

Then **double** the whole thing. Switchboards, chillers, batteries, generators and corridors need roughly as much space as the computers do.

THE RULE

3 m² per rack, then double it

DO

FROM STEP 03

rack count × 3 m²

=

M² COMPUTER ROOM

× 2 for everything else

=

M² BUILDING

QUESTION

Look around the room you are sitting in.

Is your facility bigger or smaller than this one? Pace it out if you need to.

STEP 07 OF 08

What is the bill that never stops?

GUIDE

The building runs every hour of every day. There are **8,760** hours in a year, and none of them are free.

Use **USD 0.10** per kilowatt hour unless you know your local price, in which case use that instead and say so.

THE RULE

8,760 hours a year, USD 0.10 per kWh

DO

FROM STEP 05

building power × 8,760

=

KWH PER YEAR

× USD 0.10

=

USD PER YEAR

QUESTION

That figure arrives every year, before anyone is paid and before a single server is replaced.

What else could your university buy, every year, for that amount?

STEP 08 OF 08

And the cheque you write once?

GUIDE

Building a data centre costs roughly **USD 10 million per megawatt** of computing capacity. That covers the shell, the power equipment, the cooling plant and the fit-out.

Note carefully: per megawatt of **computing**, not per megawatt of building. Use your step 02 answer, not your step 05 answer.

THE RULE

USD 10 million per MW of computing

DO

FROM STEP 02

computing load ÷ 1,000

=

MW

× USD 10,000,000

=

USD, ONCE

QUESTION

Take your step 07 answer and multiply it by ten.

Over a ten year life, which costs more: the building, or the electricity that runs through it?

Eight answers. Do they look right?

Check before you go on. If one is far off, go back to that step rather than carrying the error into your card.

STEP	YOUR NUMBER	SHOULD BE ROUGHLY
01	Server power	200 kW
02	Computing load	220 kW
03	Racks	20
04	Power per rack	Around 11 kW, under the limit
05	Building power	330 kW
06	Building size	100 to 150 m²
07	Annual electricity	USD 250,000 to 300,000
08	Cost to build	USD 2 million to 3 million

The two places people slip: adding the 10% to the wrong number at step 2, and using the building power instead of the computing load at step 8.

STEP 09 OF 10

How reliable does it need to be?

GUIDE

Reliability is bought by having spares that sit idle waiting for a bad day. A second power feed, a spare chiller, a duplicate of everything.

Each level up costs more to build **and** more to run, because you are powering equipment that does nothing most of the time.

Basic	One power supply, one cooling system	about 29 h
Better	Spare parts, still one path into the building	about 22 h
Good	Two of everything, service without stopping	about 1.6 h
Best	Two live systems, survives a fault mid service	about 26 min

DO

Circle one level:

BASIC / BETTER / GOOD / BEST

Then write one sentence saying why it is right for a university research facility.

QUESTION

Research jobs can be restarted. Surgery cannot.

Justify your choice against a consequence, not against ambition. "Because it is the best one" is not an answer.

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STEP 10 OF 10 • YOUR TEAM DRAWS ONE CARD, AND NOBODY ELSE HAS YOURS

CARD A • THE AI GROUP HAS BEEN FUNDED

They want to add **20 GPU servers**. Each draws **10 kW**, and only **4 fit in a rack**.

- How much power do the GPU servers add on their own?
- What is the new computing load, and the new building power?
- What is the power per rack in the GPU racks?
- Compare that to step 4. Can the room you designed cool them?

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Tell the class in one sentence what has to change, and roughly what it does to the annual bill.

CARD B • THE EFFICIENCY QUESTION

An architect says she can deliver the same facility at a ratio of **1.2** instead of 1.5, for an extra **USD 400,000** to build.

- Redo step 5 and step 7 with 1.2 instead of 1.5
- How much does she save you each year?
- Over ten years, how much is that?
- Is the extra USD 400,000 worth spending?

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Tell the class the year in which the better building has paid for itself.

CARD C • THE POWER COMPANY SAYS NO

The campus substation can supply the building with **250 kW** in total, and an upgrade takes two years.

- Compare that to your step 5 answer. By how much are you over?
- Working backwards through steps 5, 2 and 1, how many servers actually fit inside 250 kW?
- Now try it again with a ratio of 1.2 instead of 1.5. How many fit then?

Tell the class how many servers good engineering just bought you, without any help from the power company.

CARD D • THE SUSTAINABILITY QUESTION

Every kilowatt hour on this grid produces about **0.5 kg of CO2**.

- Take your step 7 answer in kWh. How many tonnes a year is that? (1,000 kg = 1 tonne)
- An average petrol car produces roughly 4 tonnes a year. How many cars is your building?
- On a cleaner grid, at 0.05 kg per kWh, what happens to the number?

Tell the class one non-technical reason a university might build here anyway.

ONE CARD, COPIED FROM YOUR EIGHT ANSWERS. WRITE IT BIG ENOUGH TO HOLD UP

Step 02 · computing load	KW
Step 03 · racks	COUNT
Step 04 · power per rack	KW · AIR COOLED? Y / N
Step 05 · building power	KW
Step 06 · building size	M ²
Step 07 · annual electricity	USD PER YEAR
Step 08 · cost to build	USD ONCE
Step 09 · reliability, and why	ONE SENTENCE
Step 10 · our card, and the answer	ONE SENTENCE

WHEN YOU PRESENT

Ninety seconds. Lead with your **step 05 answer**, the building power in kilowatts.

That is the number this industry leads with, and by tomorrow you will know why.

WHAT TO TAKE AWAY

A data centre is a power station with **computers** attached.

THE REAL LIMIT

Space is rarely the constraint. Power almost always is

THE REAL COST

The building arrives once. The bill arrives every year

WHY EFFICIENCY

It decides whether you fit inside your supply at all

HONESTY

Every number today is a rule of thumb. Real projects argue for months